

Adnan Ahmad

Machine Learning Researcher — Multimodal AI, LLMs & Distributed Learning

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PROFESSIONAL PROFILE

Machine Learning researcher with a PhD and 13+ peer-reviewed publications, specialising in robust and adaptive learning for complex, real-world environments. Research expertise spans federated and decentralized learning, multimodal AI, and reinforcement learning, including the development of robust aggregation methods for heterogeneous data distributions using second-order loss-landscape information and differential geometry. More recently, focused on multimodal human-AI systems that leverage physiological and behavioural signals, including EEG, ECG, and human intent, to develop adaptive policies for efficient and robust human-AI coordination in dynamic and high-stakes settings. Current interests include large language models (LLMs), with a focus on parameter-efficient adaptation of federated foundation models and retrieval-augmented generation (RAG) systems.

RESEARCH AREAS

- **Large Language Models and Generative AI:** Foundation-model adaptation, parameter-efficient fine-tuning (PEFT), federated LLM adaptation, and retrieval-augmented generation (RAG).
- **Robust and Federated Machine Learning:** Federated and decentralized learning, learning under heterogeneous and non-IID data distributions, robust aggregation, and adaptation under distribution shift.
- **Multimodal AI and Human-AI Interaction:** Multimodal deep learning, representation learning, physiological signal modelling (EEG and ECG), behavioural intent modelling, and human-AI coordination for intelligent systems.
- **Continual Learning and Adaptive Decision-Making:** Continual and self-supervised learning, multi-objective reinforcement learning, and adaptive AI systems for dynamic and non-stationary environments.

EDUCATION

Doctor of Philosophy in Information Technology (Machine Learning)

Dec 2020 – Sep 2024

Deakin University

Burwood, VIC, Australia

Thesis: Robust and Communication-Efficient Federated Learning under Non-IID Data. Developed optimization and aggregation algorithms for privacy-preserving federated learning under statistically heterogeneous data, addressing communication-efficient distributed optimization, scalable deep learning, and decentralized learning via second-order optimization and representation learning on benchmark computer-vision datasets.

Bachelor of Science in Electrical Engineering (Computer Engineering)

Sep 2013 – Sep 2017

COMSATS University

Islamabad, Pakistan

Final Year Project: FPGA implementation of general-purpose processors and hardware accelerators for computer-vision applications, compared on throughput-to-area ratio.

RESEARCH EXPERIENCE

Associate Research Fellow – Applied Machine Learning & Human-AI Systems

March 2024 – May 2026

Deakin University

Waurin Ponds, VIC, Australia

- Conducted research on an AOARD-funded project developing multimodal machine learning methods for human-AI interaction, integrating EEG, ECG, eye-tracking, and behavioural signals to model human states and enable robust, adaptive decision-making.
- Developed deep, continual, self-supervised, and representation-learning methods to improve cross-subject generalization and robustness under distribution shift in multimodal physiological data.
- Developed adaptive policies using multi-objective reinforcement learning to support efficient and robust human-AI coordination in dynamic and high-stakes environments.
- Built real-time multimodal AI systems for continuous physiological data acquisition and inference using Lab Streaming Layer (LSL), Docker, and Google Cloud Platform, with low-latency processing pipelines.
- Co-supervised PhD and Master's students, contributed to research ethics applications and competitive grant proposals, and supported research dissemination and outreach activities.

Machine Learning Researcher (Decentralized Federated Learning)

July 2024 – Jan 2025

National Research Council of Italy (IIT-CNR) — ERCIM Fellowship

Pisa, Italy

- Developed privacy-preserving optimization and aggregation algorithms for heterogeneous distributed AI systems.
- Designed a second-order (Hessian-informed) decentralized aggregation framework that improved convergence and communication efficiency under non-IID data, without a central parameter server.
- Implemented scalable distributed-learning frameworks in PyTorch, evaluated across heterogeneous peer-to-peer network topologies on benchmark computer-vision datasets.
- Collaborated with an international team on algorithm development, large-scale experimentation, publications, and open-source research software.

Casual Academic (Teaching Assistant — Machine Learning)

March 2021 – March 2024

Deakin University

Burwood, VIC, Australia

- Delivered workshops in deep learning, machine learning, and computational intelligence, covering computer vision, NLP, neural networks, optimization, and intelligent systems.

PUBLICATIONS

Full list: [Google Scholar](#)

Journal Articles

1. A. Ahmad, W. Luo, and A. Robles-Kelly, “Robust federated learning under statistical heterogeneity via Hessian-weighted aggregation,” *Machine Learning*, vol. 112, 2023.
2. A. Ahmad, W. Luo, and A. Robles-Kelly, “Robust federated learning under statistical heterogeneity via Hessian spectral decomposition,” *Pattern Recognition*, vol. 141, 2023.
3. A. Ahmad, B. Nakisa, and M. N. Rastgoo, “Robust emotion recognition via bi-level self-supervised continual learning,” *Neurocomputing*, 2026.
4. A. Ahmad, B. Nakisa, and M. N. Rastgoo, “Adaptive prototype aggregation for continual learning on physiological data under cross-subject variability,” *ACM Transactions on Intelligent Systems and Technology*, 2026.
5. S. R. Naqvi, A. Ahmad, S. M. R. Islam, et al., “Towards prevention of sportsmen burnout: formal analysis of sub-optimal tournament scheduling,” *Computers, Materials & Continua*, vol. 70, 2022.
6. A. Ahmad, C. Boldrini, L. Valerio, A. Passarella, and M. Conti, “DecHW: Heterogeneous decentralized federated learning exploiting second-order information,” *IEEE Transactions on Emerging Topics in Computational Intelligence*, under review.
7. A. Ahmad, B. Nakisa, and M. N. Rastgoo, “Partner-aware hierarchical skill discovery for robust human-AI collaboration,” *Expert Systems with Applications*, under review.
8. A. Ahmad, B. Nakisa, and M. N. Rastgoo, “Adaptive human-AI coordination via hierarchical action disentanglement,” *Transactions on Machine Learning Research*, under review.
9. A. Ahmad et al., “Bio-inspired dual-network model to tackle statistical heterogeneity in federated learning,” 2023.
10. A. Ahmad, W. Luo, and A. Robles-Kelly, “Federated learning under statistical heterogeneity on Riemannian manifolds,” in *PAKDD*, 2023.
11. A. Ahmad, W. Luo, and A. Robles-Kelly, “A Hessian-based federated learning approach to tackle statistical heterogeneity,” in *Advanced Data Mining and Applications*, Springer, 2023.
12. N. Fernando, B. Nakisa, A. Ahmad, and M. N. Rastgoo, “Adaptive XAI in high-stakes environments: modeling swift trust with multimodal feedback in human-AI teams,” 2025.
13. D. Spina et al., “Report on the 3rd workshop on neurophysiological approaches for interactive information retrieval (NeurophysIIR 2025) at ACM SIGIR CHIIR 2025,” 2025.

REVIEWER ACTIVITIES

Reviewer for NeurIPS (2026), ICLR (2025), Elsevier *Pattern Recognition*, Elsevier *Pervasive and Mobile Computing*, PAKDD, IEEE IJCNN, and the WACV Workshop on Complex-Valued Deep Learning / SARFish Challenge.

AWARDS & RECOGNITION

- ERCIM “Alain Bensoussan” Fellowship Programme — Offered 2024
- Postgraduate Research Scholarship, Deakin University — 2020–2024
- Chartered Professional Engineer (CPEng), Engineers Australia
- Registered Engineer, Pakistan Engineering Council (PEC)
- Deakin Cyber Research Grant — AUD 12,000

TECHNICAL SKILLS

Machine Learning: Federated & Decentralized Learning, Multimodal Learning, Foundation Models, LLMs, Parameter-Efficient Fine-Tuning, Continual / Self-Supervised / Representation Learning, Reinforcement Learning

Programming: Python, SQL, C++ , Java, Verilog

Frameworks: PyTorch, TensorFlow, Hugging Face Transformers, LangChain, SentenceTransformers, Scikit-learn, Pandas, NumPy, OpenCV

LLM & Retrieval: RAG, Vector Databases (Qdrant), Prompt Engineering

Cloud & MLOps: Docker, FastAPI, Git, MLflow, Weights & Biases, GCP (Compute Engine, Vertex AI), LSL

Distributed & Hardware: Peer-to-Peer Training, HPC, CUDA, NVIDIA Jetson Nano, FPGA (Xilinx Virtex-5)

SELECTED PROJECTS

- **InsightDocs — Production Agentic RAG** — Deployed retrieval-augmented generation system with hybrid retrieval (dense + BM25) and cross-encoder reranking, an agentic router and query decomposition, and a quantitative evaluation harness; serves grounded, cited answers via FastAPI + Qdrant + Docker. [Project](#) | [GitHub](#)
- **MedVision — Explainable Skin-Lesion Classifier** — Fine-tuned a ConvNeXt classifier on DermaMNIST (7 classes) to 0.91 macro-AUROC with Grad-CAM explanations; optimized inference via ONNX + INT8 (3.4× faster, 3.9× smaller), served through a Dockerized FastAPI app with CI. [Project](#) | [GitHub](#)
- **Decentralized Federated Learning with Second-Order Optimization** — PyTorch peer-to-peer FL framework with a Hessian-based aggregation method (DecHW) that outperforms FedAvg on convergence and communication cost under non-IID data. [Project](#) | [GitHub](#)
- **Real-Time Human–AI Coordination via Physiological Feedback** — Overcooked system where models infer human mental state from EEG/ECG/eye-tracking and RL agents adapt; built on LSL streaming, Docker, and FastAPI. [Project](#) | [GitHub](#)
- **Real-Time Conversational AI for Pepper Robot** — Low-latency speech-to-response system integrating a SoftBank Pepper robot with GPT via an API-driven architecture. [Project](#) | [GitHub](#)
- **Pipelined MIPS Processor & Vision Accelerators on FPGA** — Pipelined MIPS processor and low-area vision accelerators (Verilog) on Xilinx Virtex-5, plus a Java assembler, for hardware–software co-design and latency/area/throughput analysis. [Project](#) | [GitHub](#)

OUTREACH

Student Ambassador, Deakin University (2022–2024) — delivered AI and robotics demonstrations to large audiences during Open Day.